

GOVERNMENT ENGINEERING COLLEGE KISHANGANJ

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COMPUTER NETWORKS LAB-02

Exploring Packet Sniffer and Packet Analyzer

Analysis source: the submitted Wireshark .pcapng capture. Values below are taken from the observed packets in that capture.

Objective

To study network packet structures and identify protocols and header fields at different layers using Wireshark, and to analyze HTTP/HTTPS traffic generated while accessing NeverSSL and Cornell University.

Observed Capture Details

Item	Observed value
Client IPv4 address	10.202.158.13
NeverSSL IPv4 address	34.223.124.45
Client IPv6 address	2409:40e5:1171:d9f2:88a2:4d7a:bffd:e02f
Cornell IPv6 endpoint observed	2603:1061:14:151::1
NeverSSL HTTP connection	TCP 63460 -> 80
Cornell HTTPS connection (initial observed access)	TCP 50035 -> 443

Q1. Protocols observed at different layers

Layer	Protocols observed
Application Layer	HTTP, HTTPS/TLS, DNS
Transport Layer	TCP, UDP
Network Layer	IPv4 and IPv6

NeverSSL generated unencrypted HTTP traffic on TCP port 80, so its GET request and HTTP response were directly visible. Cornell was accessed using HTTPS on TCP port 443; therefore, the application data was carried as encrypted TLS traffic.

Q2. Total amount of data received

Website	Server -> client frames	Captured frame bytes	TCP/Application payload bytes
http://neverssl.com/	6	4,633 bytes	4,305 bytes HTTP/TCP data
https://www.cornell.edu/	39	92,056 bytes	89,158 bytes encrypted TCP/TLS payload

Note: 'Captured frame bytes' includes link/network/transport headers. For NeverSSL, the HTTP response itself contains 4,305 bytes of TCP payload (HTTP headers + 3,961-byte response body). For Cornell, HTTPS is encrypted, so the capture shows encrypted TLS/TCP payload rather than readable HTTP response content.

Q3. Analysis of <http://neverssl.com/>

A. Number of HTTP GET requests

For the requested NeverSSL access generated from the terminal, one HTTP GET request was observed:

GET / HTTP/1.1

The capture also contains later NeverSSL-related browser/background traffic (/online, /online/, and /favicon.ico). These are separate from the terminal access used for the main NeverSSL measurement.

B. HTTP version used by the server

The server response was: HTTP/1.1 200 OK. Therefore, the observed server response uses HTTP/1.1.

C. TCP segments and HTTP response data

GET request	HTTP response	TCP data segments received	Total HTTP/TCP response data
GET / HTTP/1.1	HTTP/1.1 200 OK	3 unique data-carrying TCP segments	4,305 bytes

The response header reports Content-Length: 3961 bytes. The complete TCP-carried HTTP response is 4,305 bytes, consisting of the HTTP headers plus the 3,961-byte response body. The response data is carried in three unique data-bearing TCP segments (2,600 bytes, 1,300 bytes, and 405 bytes).

D. TCP three-way handshake

The TCP connection for the terminal NeverSSL request was established between client 10.202.158.13:63460 and server 34.223.124.45:80.

Packet No.	Direction	Flags	Raw Seq.	Raw Ack.	Relative Seq.	Relative Ack.
574	Client -> Server	SYN	586161769	0	0	0
639	Server -> Client	SYN, ACK	2630528623	586161770	0	1
640	Client -> Server	ACK	586161770	2630528624	1	1

Thus, the handshake is SYN -> SYN-ACK -> ACK. Wireshark normally displays relative TCP sequence numbers by default, so the visible relative sequence numbers are 0, 0, and 1 respectively. The raw sequence numbers are included above for verification against the capture.

Conclusion

The experiment demonstrates how Wireshark captures packets at different protocol layers. DNS resolves host names, TCP establishes reliable connections using the three-way handshake, HTTP traffic to NeverSSL can be inspected in plain text, while Cornell HTTPS traffic is protected by TLS and appears encrypted in the packet capture.